# Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
$4^{th}$ Year $1^{st}$ Semester Final Examination, 2024
Course Code: EEE 421, Session: 2020-2021
Course Title: Solid State Devices

**Full Marks: 100** **Credit: 3.0** **Total Time: 3 Hours**

## Part-A

*[Answer any **two** of the following questions]*

1. (a) Sketch the position of Fermi level in Si at 300 K as a function of doping concentration. 5

   (b) How do you define a minority carrier? Why does it diffuse? What is Poisson's equation and why is it important in PN junction? 5

   (c) Determine the number of quantum states($cm^{-3}$) in silicon between $E_c$ and $E_c+kT$ at $T = 300$ K. 5

   (d) A Si sample is doped with $N_d = 1 \times 10^{15}$ $cm^{-3}$ of As atoms. Calculate the electron and hole concentrations and the position of Fermi level position (w.r.t. $E_c/E_v$) at 300 K, assuming full ionization. Justify the assumption by calculating the probability of doner [*sic*: should be donor] states being occupied by electrons and therefore not ionized (assume, $E_C - E_D = 50$ meV). Repeat the above calculations for $T = 30$ K and comment on the values and assumptions. [$N_c = 2.8 \times 10^{19}$ $cm^{-3}$, for Si]. 10

2. (a) Draw energy band diagram of a compensated semiconductor. 5

   (b) Derive the Einstein relation relating the mobility and diffusion constant. Also, discuss the significance of the Einstein relation. 5

   (c) Using the expression for $V_{bi}$, find the width of the depletion region.

   $$W = \left\{ \frac{2\epsilon_s V_{bi}}{e} \left[ \frac{N_a + N_d}{N_a N_d} \right] \right\}^{\frac{1}{2}}$$ 5

   (d) Answer the following questions on Fermi level: 10

   i. At what energy level above Fermi level (in unit of $kT$) the probability of finding an electron is 10% ?

   ii. Find the electron concentration($n_0$) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature, $N_c = 2.8 \times 10^{19}$ $cm^{-3}$).

3. (a) Starting from charge neutrality condition $n_0 + N_a^- = p_0 + N_d^+$, derive the expressions for $n_0$ and $p_0$ in terms of $N_a$, $N_d$, and $n_i$. 5

   (b) Sketch Energy-band diagram of a PN junction in thermal equilibrium and show P region, N region, $E_C$, $E_V$, $E_F$, $E_{Fi}$, $e\phi_{Fn}$, $e\phi_{Fp}$ and $eV_{bi}$. 5

   (c) Define drift mobility. Explain the mechanism of carrier scattering in semiconductor materials and significance of $\tau$. 5

   (d) For the following questions: 10

   i. Solve the ambipolar transport equation for a semiconductor with $N_a = 5 \times 10^{16}$ $cm^{-3}$ for $x > 0$ at steady state with zero electric field where excess charges are created only at $x = 0$. ($\delta n_0 = 5 \times 10^{15}$ $cm^{-3}$).

   ii. Find the distance where the excess charge will drop to $10 \times 10^{13}$ $cm^{-3}$. (Assume $\tau_{n0} = 7 \times 10^{-7}$ s, $\tau_{p0} = 5 \times 10^{-7}$ s, $D_n = 30$ $cm^2$ $s^{-1}$, $D_p = 25$ $cm^2$ $s^{-1}$)


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iii. Plot n and p as function of x in log scale.

## Part-B

*[Answer any **two** of the following questions]*

4. (a) Explain the effects of temperature on carrier mobility inside a semiconductor. 5

   (b) Draw the energy band diagram of a Metal-Oxide-Semiconductor structure for the three modes of operation of MOSFET and thus explain how the channel is formed. 5

   (c) Show the four operating modes of a bipolar transistor as a function of $V_{BE}$ and $V_{CB}$. Draw the minority carrier distribution of the four modes of a NPN transistor with proper labelling of all the quantities. 5

   (d) Consider a PN junction at room temperature with $n_i = 1.5 \times 10^{10}$ cm$^{-3}$, assume doping in N side is $1 \times 10^6$ cm$^{-3}$ [*sic*: should be $1 \times 10^{16}$ cm$^{-3}$]. What will be the minority carrier hole concentrations at the edge in N side if forward bias and reverse bias of 0.7 V is applied? Do the same for majority carrier electron concentration in N side. 10

5. (a) Write the ambipolar equation for $g' = 0$ for n type semiconductor with constant electric field $E_0$. At time $t = 0$, a number of electron-hole pair is generated at $x = 0$. Sketch excess electron concentration as a function of distance at different time. 5

   (b) For NPN transistor, plot minority carrier concentration in base, emitter, and collector as a function of space. What are the minority carrier concentrations at the boundaries of base, emitter, and collector? 5

   (c) Explain the working principle of Tunnel diode. Draw its characteristics curve. 5

   (d) Current is flowing through an extrinsic semiconductor placed in a magnetic field as shown in Fig. 1. 10

      i. What is the Hall voltage?
      ii. What is type of carrier?
      iii. What is the carrier concentration?
      iv. What is the mobility?

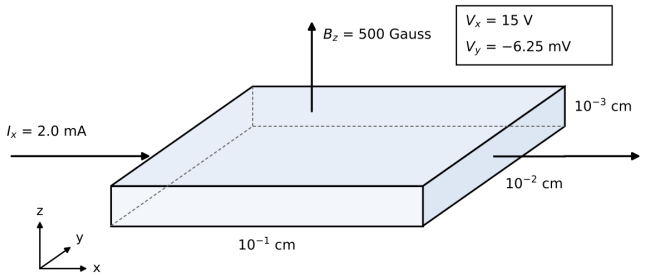



Fig. 1: A bar of an extrinsic semiconductor in a magnetic field (redrawn from the paper)

6. (a) Draw the schematic of a double-gate MOS structure. What are the benefits of such structure over single-gate MOS? 5

   (b) Briefly explain the terms complete ionization and freeze out. Can we achieve complete ionization in room temperature? Prove your answer. 5

   (c) Explain production of inverse layer at oxide and semiconductor layer with proper sketches of energy band diagram. 5


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(d) A silicon PN junction at $T = 300$ K with zero applied bias has doping concentration of $N_d = 5 \times 10^{16}$ cm$^{-3}$ and $N_a = 5 \times 10^{15}$ cm$^{-3}$. Determine built-in potential, space charge width and electric field at metallurgical region ($\epsilon_t = 11.7$ [*sic*: should be $\epsilon_r = 11.7$]) 10


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